SmartAg Ops Advisor — Market Requirements Document v3.2

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SmartAg Ops Advisor helps a small tomato grower near Oddanchatram see, per 14 kg crate, which nearby market leaves the most money after picking and selling costs, and says plainly when the data cannot support an answer.

Headline insight: in Dindigul's March 2026 crash, the real question was not "which market pays most?" but "will any market cover the cost of picking this batch?"

Version 3.2 (consolidates v3.0 and v3.1) · Pilot: tomato, Dindigul–Oddanchatram (provisional) · Evidence grades: A official or peer-reviewed, B credible news, C assumption or one interview.

1. Problem and evidence

A higher reported price can leave a small farmer with less once picking, transport and selling costs are paid, and there is no quick way to check this per crate before spending that money.

Evidence	What it shows	Grade
Dindigul, Mar 2026 [1]	Box fell to Rs 100–150 against about Rs 80 picking; harvest halted; Rs 20–70 left before transport	B
Kolar, Mar 2025; Oddanchatram, Aug 2024 [2][3]	Harvest plus transport exceeded price; prices vary by quality and crate size (14 vs 15 kg)	B
RBI WP 08/2024; Agriculture Census 2015–16 [4][5]	Farmers get about 33% of the consumer rupee; 92.5% of TN holdings are small or marginal	A
Mitra et al. 2018 [6]	Price information changed pass-through, not average prices: a decision aid, not an income promise	A

2. Users and need

Primary persona, Murugan (working persona, Grade C): 42, 1.5 acres of tomato near Oddanchatram, sells 40–60 crates (14 kg) per harvest, basic Android phone, prefers Tamil. His question: "the other market quotes more, but is it worth the lorry?" Today he calls two traders and checks WhatsApp rate photos: undated, mixed units, never net of costs.

Need: as a small tomato grower with one batch, I need to see, per crate, what each reachable market leaves me after picking, transport and selling costs, before I pay for picking or a vehicle. Supporting users (hypotheses): a cash-constrained same-day seller and an FPO coordinator pooling loads.

3. Alternatives and differentiation

Alternative	Offers	Gap for our user
Calls, WhatsApp rate boards	Trusted, same-day	Undated, mixed units, not net of costs
KhetiTools calculator [7]	Freight, commission, break-even mandi price	No crate unit, no picking cost
ACROP [8]	Dated AGMARKNET prices; flags single-mandi data	No cost-adjusted ranking
Agrimark, Uzhavan, AGMARKNET [9][10]	Official prices	Daily reports, not bids; no net view

Differentiation (hypothesis): the Harvest Cover Check (amount left per crate and the crate price needed to cover costs), with crate units, per-trip load fill and a refusal to rank weak data; claimed against the tools we tested, not as a world first.

4. Market requirements

ID	Must-have requirement	Acceptance
MR-01	Choose locality and crop	Unsupported crop: "not validated yet"
MR-02	Price provenance	Source, date, unit, type on every price
MR-03	Safe units	Kg, quintal, 14 kg crate; unknown unit not ranked
MR-04	Comparability	Wholesale and Uzhavar Sandhai not ranked together

ID	Must-have requirement	Acceptance
MR-05	Freshness	Older than 2 days (IST) excluded from ranking
MR-06	Insufficient data	Fewer than two eligible markets: no winner
MR-07	Quantity and costs	Trip cost per market; blank is not zero
MR-08	Estimate and rank	Rank only eligible options after entered costs
MR-09	Explain result	Why first, by how much, costs not included
MR-10	Separate failures	Source error differs from "no matching records"
MR-11	Label demo data	Sample prices never called live
MR-15	Harvest cover per crate	Amount left and cover price; no harvest advice

Should: MR-12 Tamil and English. **Later:** MR-13 route ETA from verified routing; MR-14 accounts, chatbot, alerts, GPS.

5. Scope, data and risks

Out of scope: forecasts, harvest advice, booking, payments, statewide claims, ML, potato. **Data status:** the 24 Oct automated collection returned no valid rows and the historical subset has no Dindigul market, so the demo uses labelled sample prices (or the 173-row 24 Oct snapshot, once its unit and coverage are confirmed).

Risks and controls: stale prices → no-rank state and dates shown; report mistaken for an offer → "verify before travel"; wrong costs → editable inputs, omissions listed; information may not raise income → no earnings claims.

6. Validation and success

Interview 5 growers and 1–2 traders (date and consent recorded), including **how they decide a batch is worth picking**. **Targets:** hand-checked cases within Rs 1; zero winners on invalid data; 4 of 5 farmers explain the ranking and name one omitted cost; 3 of 5 would check the per-crate answer before paying for picking.

Sources

[1] NIE, 22 Mar 2026 · [2] Outlook Business · [3] NIE, 9 Aug 2024 · [4] RBI WP 08/2024 · [5] Mitra et al. 2018 · [6] KhetiTools · [7] ACROP · [8] Agrimark · [9] AGMARKNET